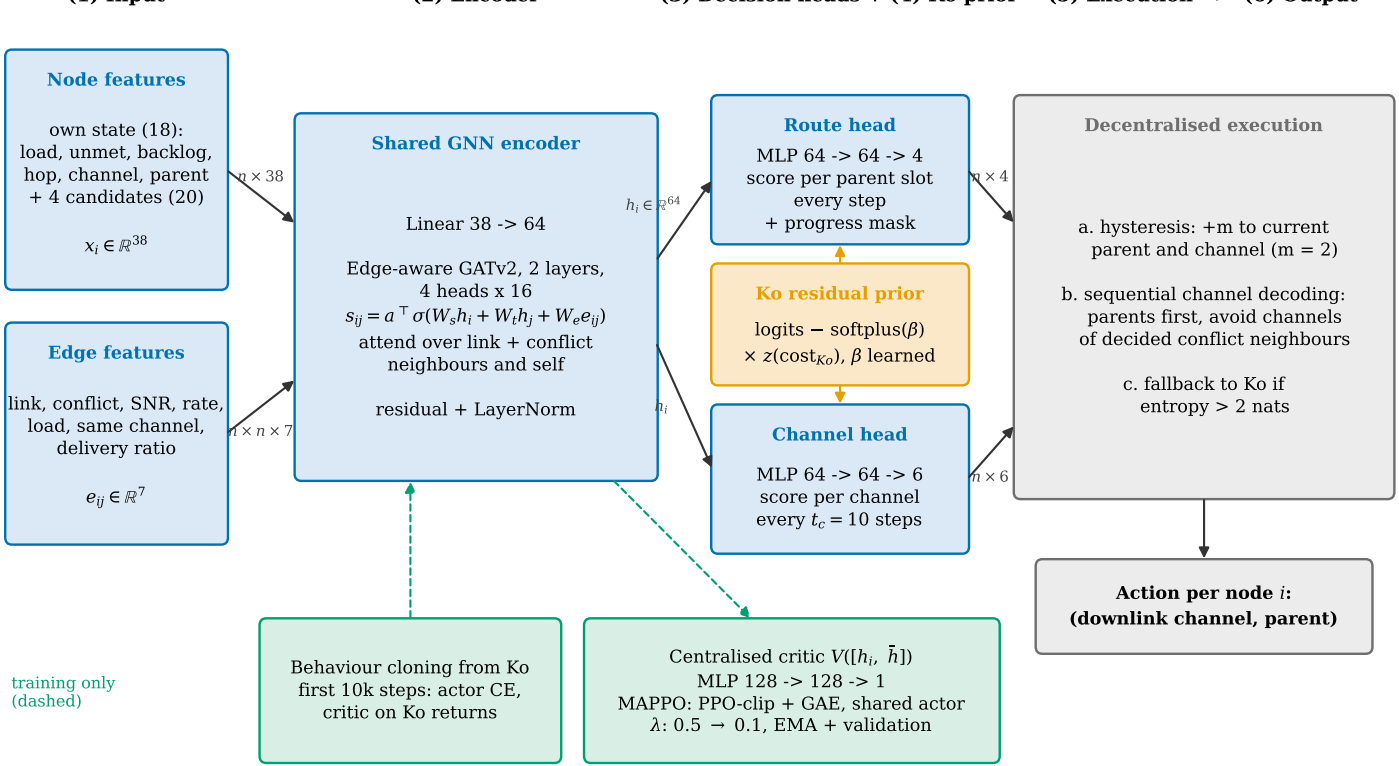
(1) Input
(2) Encoder
(3) Decision heads + (4) Ko prior
(5) Execution -> (6) Output
Node features
own state (18):
load, unmet, backlog,
hop, channel, parent
+ 4 candidates (20)
$x_i \in \mathbb{R}^{38}$
Edge features
link, conflict, SNR, rate,
load, same channel,
delivery ratio
$e_{ij} \in \mathbb{R}^7$
$n \times 38$
$n \times n \times 7$
Shared GNN encoder
Linear 38 -> 64
Edge-aware GATv2, 2 layers,
4 heads x 16
$s_{ij} = a^\top \sigma(W_s h_i + W_t h_j + W_e e_{ij})$
attend over link + conflict
neighbours and self
residual + LayerNorm
$h_i \in \mathbb{R}^{64}$
h_i
Route head
MLP 64 -> 64 -> 4
score per parent slot
every step
+ progress mask
Ko residual prior
logits − softplus(β)
× $z(\text{cost}_{Ko})$, β learned
Channel head
MLP 64 -> 64 -> 6
score per channel
every $t_c = 10$ steps
$n \times 4$
$n \times 6$
Decentralised execution
a. hysteresis: +m to current
parent and channel (m = 2)
b. sequential channel decoding:
parents first, avoid channels
of decided conflict neighbours
c. fallback to Ko if
entropy > 2 nats
Action per node i:
(downlink channel, parent)
training only
(dashed)
Behaviour cloning from Ko
first 10k steps: actor CE,
critic on Ko returns
Centralised critic $V([h_i, \bar{h}])$
MLP 128 -> 128 -> 1
MAPPO: PPO-clip + GAE, shared actor
λ: 0.5 → 0.1, EMA + validation